Data Cleaning & Validation: Pharmacy Loyalty Customer List

Michael Sillyman · Data Analyst portfolio project · Tools: LibreOffice Calc, Python · Simulated data for a 3-store US pharmacy group (Indiana & Illinois)

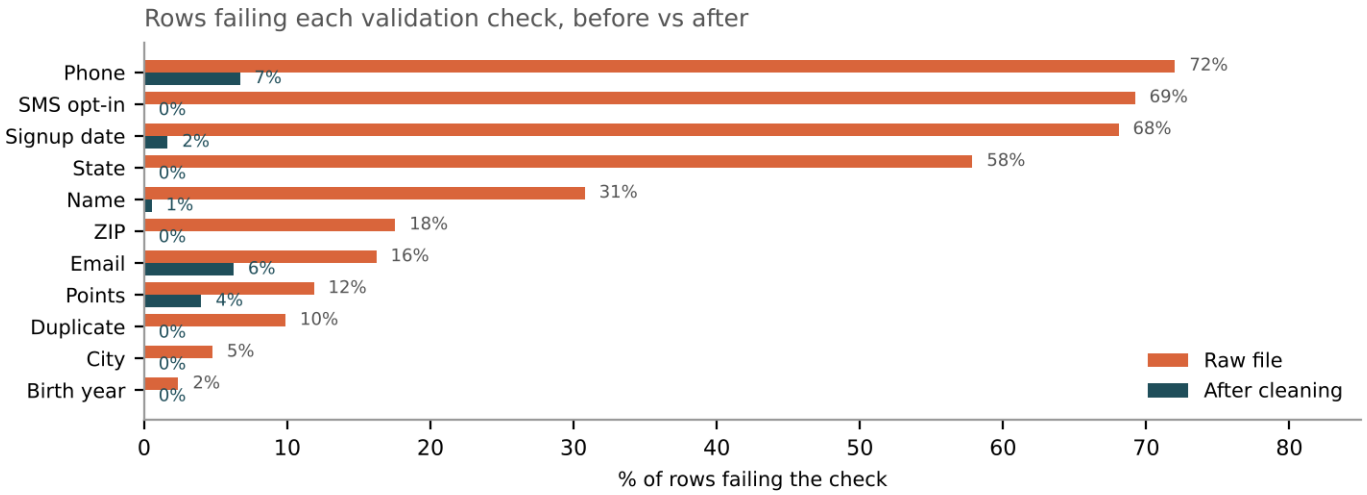
The problem

Three stores exported their loyalty customer lists and merged them into one file of 1,215 rows. Before marketing could send SMS offers or the finance team could value outstanding points, the list had to be accurate: one record per customer, contact details in a usable format, and consent recorded clearly.

Approach

1. Profile: built 11 formula checks in Calc (EXACT, PROPER, TRIM, COUNTIF, SUMPRODUCT, DATEVALUE) to measure every problem in the raw file. **2. Clean:** applied documented rules to each field, removed exact duplicates and merged customers enrolled twice, logging every change with its before and after value. **3. Validate:** re-ran the same checks on the cleaned data, and moved anything that could not be fixed from the data alone to a follow-up list.

Raw rows	Unique customers	Changes logged	Rows passing all checks	Flagged for follow-up
1,215	**1,152**	**4,275**	**0.2% → 82%**	**258**



Key findings

- **Only 2 of 1,215 raw rows met every standard.** Most problems were format inconsistencies between the three store systems: phone numbers (72% of rows), SMS opt-in coding (69%), signup dates (68%) and state names (58%).
- **63 rows were duplicates.** 35 were exact repeats of the same export; 28 were the same customer enrolled at a second store. Merged records keep the earliest enrollment and are flagged so the points balance can be confirmed.
- **Blank SMS consent was treated as "No",** not "Yes". Texting customers without recorded consent creates legal risk under US telemarketing rules (TCPA).
- **82% of cleaned records pass every check.** The rest have information the file cannot supply: 72 without a usable email, 77 without a complete phone number, 46 with a missing or negative points balance, and 19 with a signup date in the future.

Recommendations

1. **Fix data entry at the source:** use drop-down lists for state and city, a phone input mask, and a date picker in all three store systems, so formats cannot drift apart again.
2. **Make key fields required:** last name, plus at least one of email or phone, before an enrollment saves.
3. **Search before enrolling:** look up existing customers by email or phone to stop duplicate accounts.
4. **Work the follow-up list:** have staff confirm the 258 flagged records at the customer's next visit, starting with the merged and negative points balances.